INTRODUCTION

Surface chemistry is the branch of chemistry that deals with the processes occurring at interfaces between phases for example, solid and liquid, solid and gas and liquid and liquid. This topic is of immense importance to our everyday life and to numerous industries, from materials and paints to medicine and biotechnology. Surfaces play a key role in heterogeneous catalysis, formation and stability of colloids and electrode reactions. Surfaces of solids are inherently different from their bulk portion. The bonding between the atoms at the mere surface is different from that in the bulk. Hydrogen that exists in the interstellar space are formed on the surfaces of grains and dust particles. Mosquitoes and other small insects can walk on the surface of water but they will drown into the water when soaps are added in the neighbourhood. We are fascinated by the spherical shape of water droplets and mercury droplets. We are also impressed by the non-sticky wings of butterfly and leaves of plants. Blue colour of the sky and red colour of the sunset strongly attract us. In all the above only the surface of matter is important. Many of creams, lotions and other personal care products are complex emulsions. Food companies are interested in developing healthy, tasty and longlasting food products. All these are based on the principles of colloids and surface chemistry. So, surface Chemistry is an exciting topic to learn.

10.1 Adsorption and Absorption

Solid surfaces have the ability to attract the contacting species due to free valency or residual force on them.

For example: charcoal adsorbs ammonia, silica gel adsorbs water., charcoal adsorbs colorants from sugar.

These examples prove that adsorption is a surface phenomenon. In contrast to adsorption, absorption is a bulk phenomenon i.e. the adsorbate molecules are distributed throughout the adsorbent.

- Adsorbent is the material on which adsorption takes place.
- Adsorbed substance is called an adsorbate.
- The surface of separation of the two phases where the concentration of adsorbed molecule is high is known as interface.
- In adsorption, if the concentration of a substance in the interface is high, then it is called positive adsorption. If it is less, then it is called negative adsorption.
- The process of removing an adsorbed substance from the surface is called desorption.
- The gaseous molecules like He, Ne,O_2,N_2,SO_2 and NH_3 and solutions of NaCl or KCl can be adsorbed by suitable adsorbents. These are referred as adsorbates.
- Silica gel and metals like Ni,Cu, Pt, Ag and Pd and certain colloids can act as adsorbents.

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